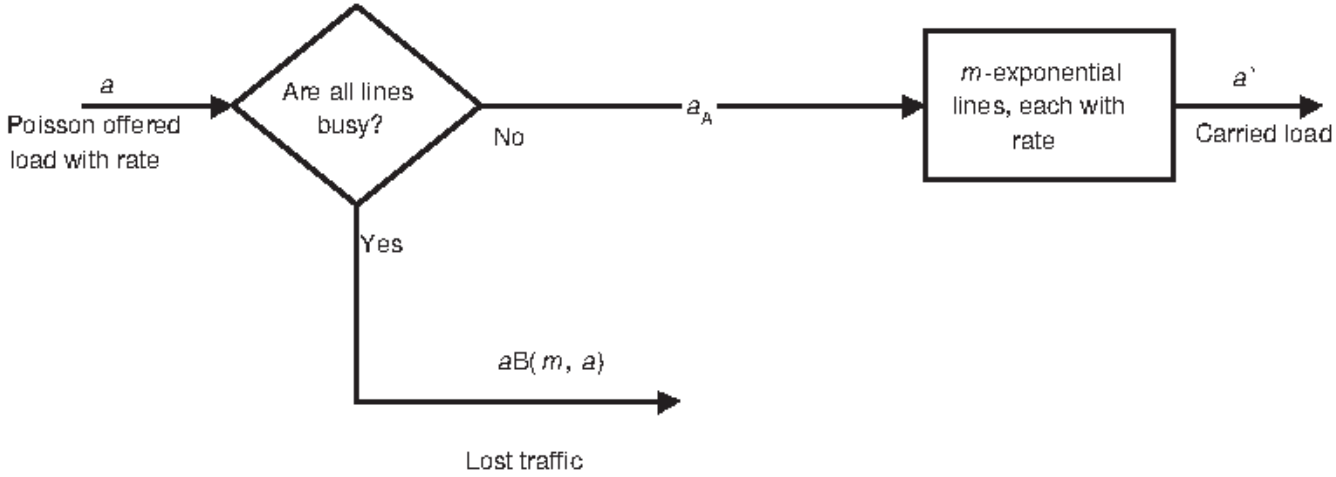
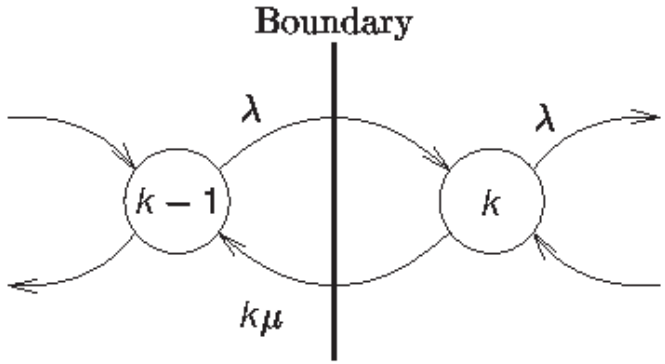
Erlang Loss

2006-10-23

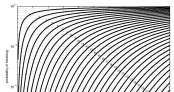


Flow chart of the Erlang Loss System.

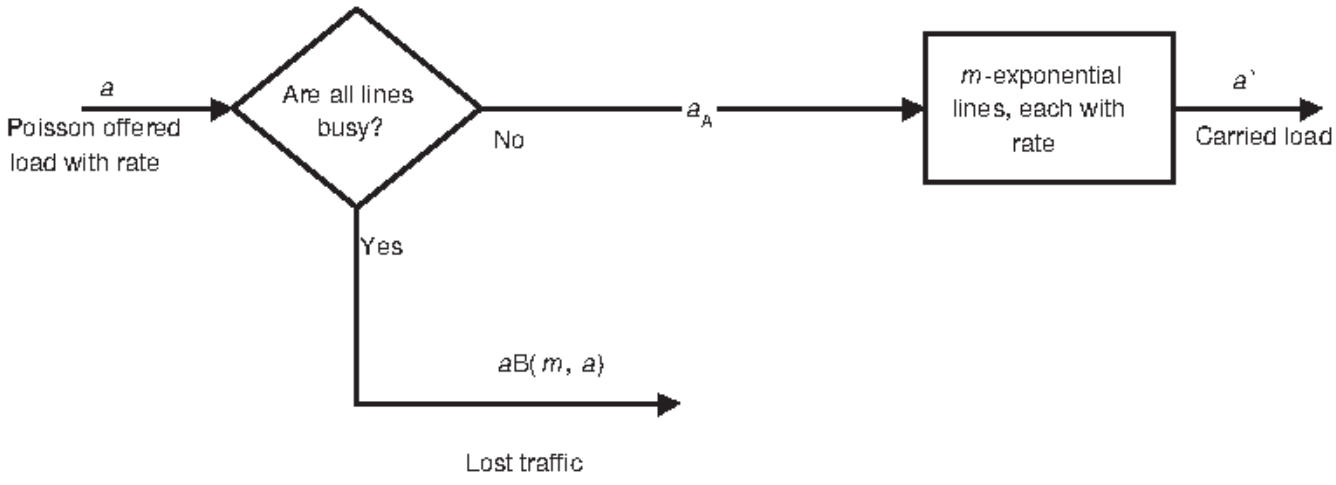
Markov chain for the Erlang loss system, $M/M/m/m$



Flow rates at a boundary.



Erlang Loss $M/M/m/m$



erlangLossFlowChart

Erlang Loss $M/M/m/m$ (continued)

- The Erlang delay system contained m servers.
- It was assumed that, if all servers were occupied, the customer would be placed on hold until a server became available.
- In the $M/M/m/m$ system there is a maximum number of customers.
-

State Diagram



erlangLoss

State Diagram (continued)

- The offered load, $a = \frac{\lambda}{\mu}$, is again dimensionless with units of Erlangs.
- The Erlang loss system may be analysed as a birth-death process with birth rate
-

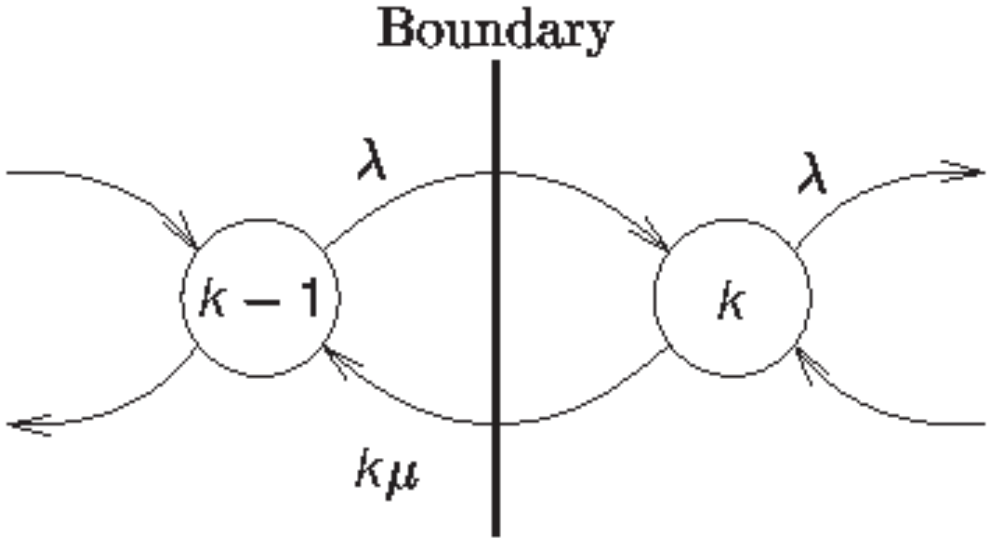
$$\mu_k = \begin{cases} k\mu & k = 0, 1, \ldots, m \\ 0 & k > m \end{cases}$$

}

Detailed Balance Equations

- When the system's state, $k \leq m$, we have boundary flow as below
- This leads to the following balance equation

$$\lambda P(k-1) = k\mu P(k).$$



boundaryFlowMMinf

Detailed Balance Equations (continued)

- So for $k = 0 \ldots m$ we may write

$$P(k) = \frac{\lambda}{k\mu} P(k-1).$$

- The system state cannot go above m so we have no need to consider the region where $k > m$. Substituting in the offered load we have,

$$P(k) = \frac{a}{k} P(k-1),$$

which means that

$$\begin{aligned} P(k) &= \prod_{n=1}^{k} \frac{a}{n} P(0) \\ &= \frac{a^k}{k!} P(0). \end{aligned}$$

Normalisation Condition

- The normalisation condition gives us

$$P(0)\sum_{k=0}^{m}\frac{a^k}{k!}=1$$

$$P(0)=\frac{1}{\sum_{k=0}^{m}\frac{a^k}{k!}},$$

- Which leads to the truncated Poisson distribution

$$P(k)=\frac{\frac{a^k}{k!}}{\sum_{n=0}^{m}\frac{a^n}{n!}}$$

- *Note if* $m\to\infty$ *we recover the* $M/M/\infty$ *queue as we would expect.*

Probability of Blocking

- For the $M/M/m$ queue we were interested in the probability that a customer would have to wait in the queue for service.
- In the $M/M/m/m$ system there is no queue because the maximum state of the system is the same as the number of servers.
- The probability of waiting is therefore zero.

Probability of Blocking (continued)

- The probability of blocking is of interest: the probability that when we arrive the system is full and we are denied service.
- As we discussed for the $M/M/1/m$ queue the blocking probability is given by $P(m)$

$$B(m, a) = P(m) = \frac{\frac{a^m}{m!}}{\sum_{n=0}^{m} \frac{a^n}{n!}}.$$

- Use Erlang B curves to compute.

Carried Load and Utilization Factors

- Since traffic is lost in $M/M/m/m$ the carried load, a', is smaller than the offered load.
- The carried load is given by the offered load multiplied by the probability that the packet is not blocked.

$$a' = a(1 - P(m)) = a(1 - B(m, a))$$

- The carried load per line is the trunk occupancy (the utilization facctor),

$$\rho = \frac{a'}{m}.$$

Carried Load and Utilization Factors (continued)

- Note the contrast with the lossless systems where

$$\rho = \frac{a}{m}$$

because the carried load is equal to the offered load $a' = a$. * This also applies for the other system with loss we have seen, the $M/M/1/m$ queue. * The rate of service in the system is no longer given by the arrival rate, λ, we need to consider those packets that are lost.

$$\dot{\lambda} = \lambda(1 - P(m))$$

- This has effects when studying the queue using Little's formula.

Load Tripling

- **Question**

Load Tripling (continued)

1. For a blocking probability of 0.01, what is the number of trunks needed for 6 Erlangs of offered load?
2. For a blocking probability of 0.01, what is the number of trunks needed for 18 Erlangs?
3. Comment on the increase in load compared to the increase in the number of trunks.

Answer

1. 13
2. 28
3. A tripling in load only leads to a (just over doubling) of the resources required. To understand this further we look at the utilization factors. For the first system the utilisation factor is

$$\rho_1 = \frac{6}{13}(1 - 0.005) = 0.459$$

whereas for the second system the utilisation factor is

$$\rho_2 = \frac{18}{28}(1 - 0.007) = 0.6384$$

The 30 trunk system is thus more efficient. This explains why it is cheaper to call the U.S.A. than Italy.

Example $M/M/m/n$

- This example is from the Exam paper in 2002-3.

Example $M/M/m/n$ (continued)

- Question

1. A call centre employs thirty operators selling tickets for a major sporting event. The call centre has sixty lines in total. When all operators are active new calls are placed in a queue for the first available operator. The call centre is to be modelled as an $M/M/m/n$ queue, *i.e.* a queue in which there are m servers and a maximum state of n.
2. Draw a state diagram representing the queueing system.
3. Write down the detailed balance equations for the two cases where the queue state $k < m$ and when $m \leq k < n$.
4. For each of the two cases outlined in the previous part write down the steady state distribution as a function of $P(0)$, the probability of the system being empty.
5. Use the fact that the steady state distribution must be normalised to show that

$$P(0) = \left[\sum_{k=0}^{m} \frac{a^k}{k!} + \sum_{k=m+1}^{n} \frac{a^k}{m! m^{k-m}} \right]^{-1}$$

Thanks!

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- The Atomic Human
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

Further Reading

- The $M/M/m/m$ loss system and Erlang B of Bertsekas and Gallager (1992)

Exercises

- Tutorial Sheet 3: trunk-group Erlang B of Lawrence (2006)

References

Bertsekas, D.P., Gallager, R.G., 1992. Data networks, Second. ed. Prentice-Hall, Upper Saddle River, NJ.

Lawrence, N.D., 2006. Erlang delay and loss and the M/G/1 queue.